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Return-to-Sport Outcomes and Psychological Readiness to Return to Sport After Revision Compared to the Primary Anterior Cruciate Ligament Reconstruction: A Systematic Review and Meta-Analysis

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Abstract

Background The existing body of evidence referring to the return-to-sport (RTS) outcomes and psychological readiness to return to sport after revision compared to the primary anterior cruciate ligament (ACL) reconstruction has not yet been comprehensively synthesized.

Objective To examine RTS outcomes and psychological readiness to return to sport after revision compared to the primary ACL reconstruction.

Methods A comprehensive search of Web of Science, EMBASE, and PubMed was carried out up to November 15, 2024. Observational studies that compared revision and primary ACL reconstruction regarding RTS outcomes and psychological readiness to return to sport were considered adequate for inclusion. The Methodological Index for Non-Randomized Studies and Grading of Recommendations Assessment, Development, and Evaluation were used to assess methodological quality as well as the overall strength and quality of evidence of involved studies, respectively.

Results A total of 19 studies fulfilled the eligibility criteria and were included in the quantitative analysis. There were 3209 athletes with a history of revision and primary ACL reconstruction (mean age: 26.0 ± 6.5 years; 78% males). The main findings indicated that individuals who were subjected to revision ACL reconstruction had slightly lower odds of RTS (-6%; odds ratio (OR) = 0.66 [95% confidence interval (CI) 0.49 to 0.89]), slightly lower odds of RTS at the preinjury level of competition (-5%; OR = 0.52 [95% CI 0.36 to 0.76]), slightly prolonged RTS time (1.2 months; standardized mean differences (SMD) = 0.38 [95% CI 0.12 to -0.63]), and moderately lower psychological readiness to return to sport (-14.2 points; SMD = -0.75 [95% CI -1.14 to -0.36]) relative to their counterparts who were subjected to the primary ACL surgery.

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Conclusions Athletes who underwent the revision ACL surgery had a 6% reduced rate of RTS, 5% lower rates of RTS at the preinjury level of competition, 1.2 months extended RTS time, and 14.2 points decreased psychological readiness to return to sport relative to the respondents with a history of the primary ACL reconstruction.

PROSPERO registration number CRD42024600347.

Key Points

- Athletes who underwent the revision anterior cruciate ligament (ACL) surgery had 6% and 5% reduced rates to return-to-sport (RTS), and to RTS at the preinjury level of competition, respectively, with 1.2 months extended RTS time compared to their counterparts with a history of primary ACL reconstruction.
- The obtained results demonstrated that psychological readiness to return to sport was 14.2 points lower after revision relative to the primary ACL surgery.
- The novel recovery approach is recommended for athletes with revision ACL reconstruction, consisting of exercise sessions and psychological interventions to improve examined parameters and prevent additional knee injuries.

Keywords Exercise psychology, Knee injury, Return to play, Return to play time, Surgery

Background

The phenomenon of anterior cruciate ligament (ACL) injury has been comprehensively investigated in the relevant body of scientific knowledge referring to sports science, sports medicine, and particularly sports orthopedics [1, 2]. Moreover, it was highlighted that ACL has a crucial role in the stabilization and kinematics of the knee joint [3]. Available evidence indicates that ACL tear is the most frequently recorded knee injury among the physically active population, participating in sports such as soccer, basketball, alpine skiing, and handball [4, 5]. Additionally, several studies reported that more than 250,000 ACL injuries per year occurred in the USA, and approximately 100,000 reconstructions were performed [6]. The available literature revealed a 1.8% ACL graft failure rate 2 years following the primary reconstruction procedure [7] and 6.2% during a follow-up period of more than 10 years [8]. There are several quite relevant predictors of the revision ACL reconstruction, including younger age at the time of surgery, participation in sports at the competitive level, and double-bundle ACL reconstruction [9].

Return-to-sport (RTS) outcomes encompass three relevant variables, such as overall RTS rate, RTS at the preinjury level of participation, and RTS time. RTS parameters have been extensively examined among individuals who were subjected to ACL reconstruction [10–15]. More specifically, Ardern et al. conducted a systematic review and meta-analysis that included 48 available studies and 5770 participants [11]. The most relevant findings of this investigation indicated that a total of 82% of athletes RTS following ACL reconstruction. In addition, a robust body of scientific evidence demonstrated that 65% of participants with a history of ACL reconstruction RTS at the preinjury level of competition [13]. Moreover, it is essential to highlight that 83% of elite athletes RTS at the same level of participation following ACL reconstruction [10]. Lastly, the period between surgery of the examined ligament and return to sports fields was approximately between 6 and 13 months, particularly depending on the type of team sport [10].

Psychological readiness to return to sport following ACL reconstruction is most commonly evaluated with the Anterior Cruciate Ligament Return to Sport After Injury (ACL-RSI) scale, which consists of three crucial components, including emotions, confidence, and risk appraisal [16]. The overall possible score for all three variables is between 0 and 100, with a greater score indicating a higher desire to return to sport among individuals who were subjected to ACL surgery [17]. Several meta-analyses already emphasized the importance of psychological readiness to return to sport with respect to the RTS phenomenon, including overall RTS rate [18] and RTS at the preinjury level of participation [19], following ACL reconstruction. More specifically, it was unambiguously demonstrated that athletes who successfully RTS had considerably greater ACL-RSI scores relative to the athletes who did not RTS [18]. Analogously, convincing evidence indicated that individuals who had RTS at the preinjury level of competition had substantially higher psychological readiness to return to sport compared to the participants who did not have RTS at the preinjury level of participation [19]. Moreover, Webster and Feller established cutoff values for psychological readiness to return to sport 6 months after ACL reconstruction, which predicted RTS outcomes at 12 months [20]. The authors highlighted that an ACL-RSI score ≥ 65 was an indicator of adequate recovery. In other words, athletes who achieved the mentioned levels of psychological readiness to return to sport had 3 to 4 times greater odds of RTS.

Currently, several studies have already synthesized existing evidence referring to each RTS outcome among athletes who were subjected exclusively to revision ACL

reconstruction [21–23]. For instance, Andriolo et al. revealed that 75% of athletes RTS after revision ACL surgery, while 43% of them RTS at the preinjury level of participation [22]. Most importantly, the available body of literature with respect to the RTS outcomes and particularly psychological readiness to return to sport following revision compared to the primary ACL reconstruction still needs to be comprehensively summarized. Additionally, examining all RTS outcomes and psychological readiness to return to sport following revision and primary ACL reconstruction should provide innovative insights regarding the overall rehabilitation process. Therefore, the goal of the current systematic review and meta-analysis was to determine RTS outcomes and psychological readiness to return to sport after revision compared to the primary ACL reconstruction. It was hypothesized that individuals with a history of revision ACL reconstruction would have lower odds of RTS, decreased odds of RTS at the preinjury level of participation, extended RTS time, and reduced psychological readiness to return to sport relative to the athletes who were subjected only to the primary surgery of ACL.

Methods

Search Strategy

Web of Science, EMBASE, and PubMed were comprehensively searched, taking into account only studies available in the English language. To provide an overview of recent and innovative evidence referring to the RTS outcomes and psychological readiness to return to sport after revision and primary ACL reconstruction, all mentioned databases were searched from January 2015 to October 2024. The overall search strategy consisted of an analysis of relevant systematic reviews with meta-analyses and MeSH expressions. A Boolean search syntax with operators "AND" and "OR" was implemented with keywords such as "return to sport," "return to preinjury level of sport," "return to sport time," "psychological readiness to return to sport," "revision ACL reconstruction," and "primary ACL reconstruction." The full search strategy with all employed keywords is given in Supplementary file (1) Concerning secondary sources of evidence, backward citation tracking (references of studies that fulfilled eligibility criteria) and forward citation tracking (mentions via Google Scholar) were carried out. Primary and secondary sources of existing scientific evidence were searched by three independent reviewers (VM, RR, and TT). In addition, to ensure that all relevant studies were encompassed, an updated search of databases was performed on November 15, 2024. The present study was conducted according to all recommendations available in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement [24]. The PRISMA checklist is provided in Supplementary file (2) The protocol for this systematic review and meta-analysis was prospectively registered via the International Prospective Register of Systematic Reviews (PROSPERO) databases (registration number: CRD42024600347).

Selection Process

Two reviewers (VM and RR) independently conducted a selection process of available literature. All discrepancies between reviewers were resolved via an extensive discussion until a consensus was reached. However, if it was not possible to reach an agreement between reviewers, the first investigator (MM) was consulted for clarification and to make the final decision. The overall study selection process comprised three relevant aspects, including screening of titles and abstracts, assessment of investigations sought for retrieval, and thorough analysis of full-text articles estimated for eligibility.

Eligibility Criteria

Studies were assessed for eligibility based on the PECOS (P - participants; E - exposure; C - comparison; O - outcome; and S - study design) framework for observational studies [25]. (P) examined population experienced ACL injury and was subjected to the reconstruction procedure. There were no restrictions regarding age, sex, and level of play among athletes; (E) revision ACL reconstruction; (C) primary ACL reconstruction. Most importantly, to improve the validity and accuracy of findings of the current systematic review and meta-analysis, only studies with 10 or more respondents who were subjected separately to revision as well as primary ACL surgery were included; (O) primary outcomes referred to the RTS parameters, including overall RTS, RTS at the preinjury level of competition, and RTS time, while secondary outcome was psychological readiness to return to sport; and (S) cohort, case-control, cross-sectional studies, and case series.

Conversely, editorials, conference papers, systematic reviews, meta-analyses, case reports, non-peer-reviewed journal articles, abstracts, qualitative studies, and books were excluded. Finally, studies with multiple revision ACL reconstructions also were not considered adequate for inclusion.

Data Extraction

Two independent reviewers (AV and AB) extracted data from the studies that fulfilled all PECOS criteria. The retrieved data were divided into the following categories: (1) authors, year of publication, and country; (2) study design and level of evidence; (3) sample size and demographic characteristics of respondents, such as age, sex, level of play, and type of sport; (4) surgical variables related to the ACL reconstruction types (primary and revision), follow-up period, and employed graft types. Of

note, the follow-up period pertained to the time between ACL revision reconstruction and evaluation of the primary and secondary outcomes; (5) RTS rate and RTS at the preinjury level of competition were expressed in numbers and percentages, while RTS time was expressed in months. Widely accepted definitions of RTS outcomes are already available in the literature [26]. Psychological readiness to return to sport was presented as a point value. Relevant data were extracted from the figures with Plot-Digitizer online software (www.plotdigitizer.com). Corresponding authors were contacted via email in case of missing data from the included studies. It is necessary to highlight that reviewers were not blinded to the names of the authors and journals regarding the data extraction as well as the study selection process. Discrepancies between reviewers were resolved after an online meeting and consultation with the first investigator (MM).

Quality Assessment

The Methodological Index for Non-Randomized Studies (MINORS) rated investigations involved in the quantitative analysis [27]. MINORS contains 8 items for non-comparative studies as well as 4 additional items for comparative studies. The maximum overall quality score is 16 and 24 for non-comparative and comparative investigations, respectively. Taking into account that each study examined RTS outcomes and psychological readiness to return to sport following revision relative to the primary ACL reconstruction, only comparative analysis was conducted. The cutoff scores for comparative investigations were as follows: very low quality (0–6), low quality (7–12), moderate quality (13–18), and high quality (19–24) [28]. Two reviewers (TT and AV) performed a quality evaluation of studies that satisfied all eligibility criteria. Potential disagreements between reviewers were resolved through discussion until a consensus was reached or following a consultation with the first investigator (MM).

Data Analysis

Data processing was performed using Review Manager software 5.4.1 (The Nordic Cochrane Centre, The Cochrane Collaboration, Copenhagen, Denmark). Continuous variables were presented as standardized mean differences (SMD) with a 95% confidence interval (CI), which were interpreted as trivial (< 0.2), small (0.2 ≤ SMD < 0.5), moderate (0.5 ≤ SMD < 0.8), and large (SMD ≥ 0.8) [29]. The odds ratio (OR) with 95% CI was calculated for categorical variables and interpreted as small (1.68), moderate (3.47), and large (6.71) [30]. Heterogeneity among studies was expressed as I^2 and considered as low (≤ 50%) and high (> 50%) [31, 32]. A fixed-effect model and random-effect model with Hartung–Knapp–Sidik–Jonkman adjustment were implemented regarding low and high heterogeneity among included studies, respectively. Differences between revision and primary ACL reconstruction with respect to the addressed outcomes were considered statistically significant if p-values were < 0.05. The robustness of the obtained results was checked with the sensitivity analysis technique, excluding studies with less than 20 respondents in the revision and primary reconstruction groups.

Certainty of Evidence Assessment

The overall strength and quality of evidence were estimated using the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) [33]. Evidence evaluated with the GRADE approach is commonly categorized as high, moderate, low, and very low. Considering that the present systematic review and meta-analysis involved only observational studies, the initial quality of evidence was low. The GRADE includes the following five aspects: (1) risk of bias, evidence was downgraded by one level if methodological quality in more than a half of the studies was low or very low; (2) inconsistency, evidence was downgraded by one level if a high level of statistical heterogeneity was recorded (I^2 > 50%); (3) indirectness, evidence was downgraded by one level if heterogeneity was observed with respect to the characteristics of the examined population; (4) imprecision, evidence was downgraded by one level if the upper and lower CIs for all outcomes had > 0.5 differences; and (5) publication bias, evidence was downgraded by one level due to noted asymmetry in funnel plots. Each aspect was evaluated only for the outcomes with 10 or more investigations. In contrast, the evidence was upgraded for each analyzed outcome if a large effect size was recorded, including the continuous (SMD) and categorical (OR) variables. Two reviewers (RR and AB) independently carried out a certainty of evidence assessment for all primary and secondary variables. The first investigator (MM) clarified all discrepancies between reviewers.

Results

Study Selection

A search of electronic databases and a selection of available studies are given in Fig. 1. A comprehensive literature search identified 1978 records and 876 duplicates were removed using Zotero software. Therefore, titles and abstracts of 1102 records were reviewed. Following the elimination of 990 records in the first phase of screening of existing evidence, 112 reports were sought for retrieval. Since only abstracts were available in 6 reports, a total of 106 full-text reports were assessed for eligibility. Reviewers (VM and RR) most frequently excluded reports if no comparison between revision and primary ACL reconstruction for assessed outcomes was recorded ($n = 27$). Regarding the secondary sources of evidence,

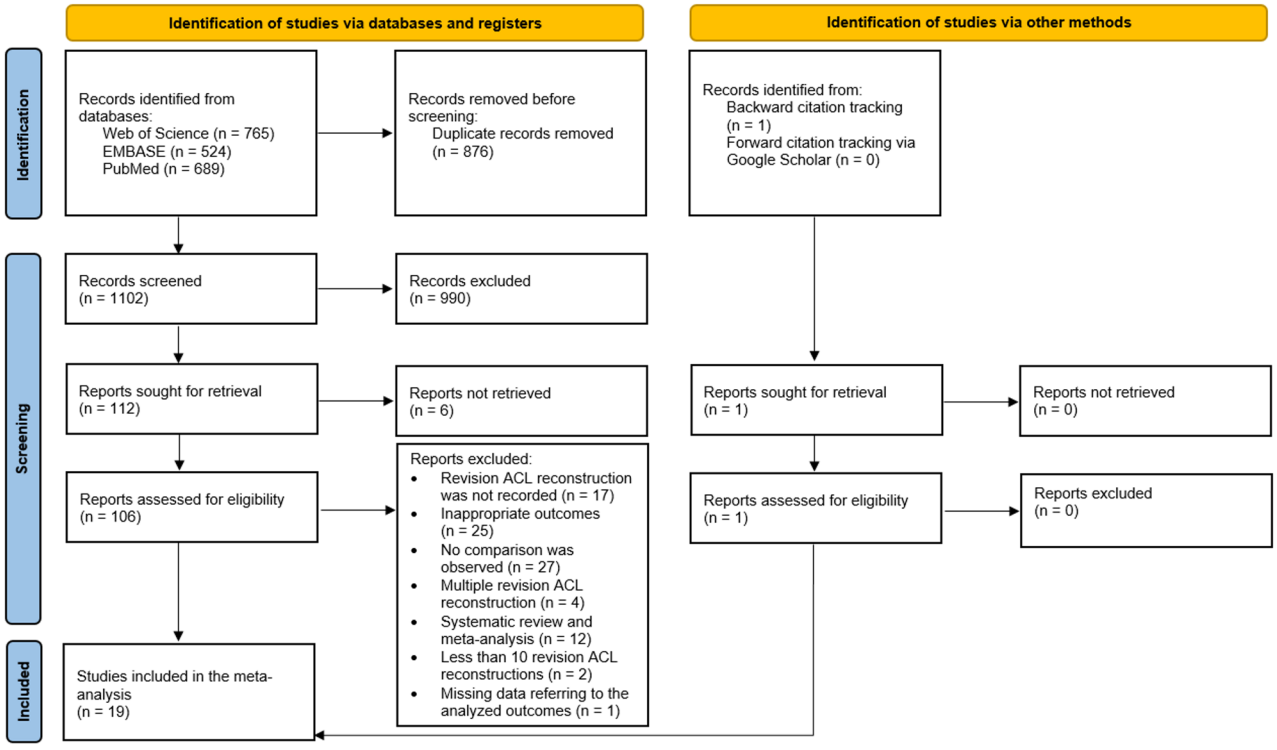


Fig. 1 PRISMA flow diagram illustrating the selection process of the included studies. ACL, anterior cruciate ligament; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses

backward citation tracking yielded 1 additional record. In summary, 19 original studies satisfied all previously emphasized PECOS criteria, and their main characteristics as well as outcomes were presented.

Study Characteristics

Studies were conducted on various continents, including Australia [34–36], Europe [37–45], North America [46–49], South America [50], and Asia [44, 51]. There were 2 prospective cohort studies [40, 43] that pertained to the level II of evidence. Further, 4 case-control [36, 45, 49, 51], 3 cohort [42, 46, 48], and 3 cross-sectional investigations [41, 44, 47] provided level III of evidence, while 7 case-series [34, 35, 37–39, 50, 52] provided level IV of evidence. A total of 3209 respondents, 2517 males and 692 females, with a mean age of 26.0 ± 6.5 years, participated in the current systematic review with meta-analysis. Most importantly, 830 athletes were subjected to revision ACL surgery, while 2736 individuals experienced only one ACL rupture and underwent the reconstruction procedure. With regard to the level of play, seven studies included exclusively elite athletes [37, 38, 45, 47–49, 52], and several investigations also solely addressed recreational [41], amateur [50], and youth athletes [36]. Studies that pertained to elite athletes quite commonly encompassed only soccer players [37, 38, 45, 47]. Elite and recreational athletes competed in 3 studies [34, 42, 43], national-level and recreational athletes participated in 2 investigations [35, 51], and one study included a sample of national-level and elite athletes [39]. Individuals with a history of revision and primary ACL reconstruction mostly competed in soccer [34–45, 47, 50], while basketball [34–36, 39, 41–44, 48], Australian Rules football [34–36, 52], and American Football [41, 48, 49] were also commonly recorded. The overall average follow-up period for RTS outcomes was 43 months (3.6 years). In addition, the mean period between ACL revision reconstruction and evaluation of psychological readiness to return to sport was 10 months. The most commonly implemented graft types were hamstring autograft and bone-patellar tendon-bone autograft (Table 1). The graft type was not reported in 5 investigations [35, 38, 45, 48, 49], including a total of 869 athletes.

Results of Syntheses

Return to Sport Outcomes

Ten studies estimated the overall RTS rate after revision and primary ACL reconstruction (Fig. 2). Individuals with a history of revision ACL reconstruction had slightly reduced odds of RTS relative to the athletes who were subjected to the primary ACL surgery (73% vs. 79%, respectively; -6%; OR = 0.66 [95% CI 0.49 to 0.89]; $p = 0.005$). A low statistical heterogeneity was observed among studies that evaluated overall RTS ($I^2 = 16\%$).

Table 1 Main characteristics of studies included in the quantitative analysis

Study	Country	Study design (level of evidence)	Population characteristics	Level of play (type of sport)	ACL reconstruction type (*n*)	Follow-up period (months)	Graft type
Anand et al. [34]	Australia	Case series (IV)	*n* = 109; M/F = 80/29; mean age = 28.0 ± 8.8 years	Recreational and elite athletes (Australian Rules football, soccer, basketball, and netball)	Revision = 109; primary = 109	58	Hamstring autograft, BTB autograft, and Achilles tendon allograft
Ardern et al. [35]	Australia	Case series (IV)	*n* = 122; M/F = 76/46; mean age = 28.3 ± 8.4 years	National-level and recreational athletes (Australian Rules football, netball, soccer, and basketball)	Revision = 18; primary = 93	24	NR
Balendra et al. [36]	United Kingdom	Case series (IV)	*n* = 232; M/F = 205/27; mean age = 23.3 ± 4.4 years	Elite athletes (soccer)	Revision = 22; primary = 231	24	Hamstring autograft, BTB autograft, and BTB allograft
Bezuglov et al. [37]	Russia	Case series (IV)	*n* = 85; M/F = 85/0; mean age = 26.2 ± 3.5 years	Elite athletes (soccer)	Revision = 11; primary = 76	144	NR
Borim et al. [38]	Spain	Case series (IV)	*n* = 19; M/F = 9/10; mean age = 29.8 ± 7.5 years	National-level and elite athletes (basketball, soccer, tennis, athletics, cycling, and rugby)	Revision = 19; primary = 19	24	BTB autograft
Carolan et al. [39]	Ireland	Prospective cohort (II)	*n* = 344; M/F = 344/0; mean age = 24.5 ± 4.3 years	NR (Gaelic Football, hurling, soccer, rugby)	Revision = 46; primary = 298	9	BTB, hamstring, and quadriceps autografts
Duncan et al. [40]	USA	Cohort (III)	*n* = 184; M/F = 110/74; mean age = 29.9 ± 10.0 years	NR (NR)	Revision = 92; primary = 92	9	BTB, quadriceps tendon, and hamstring autografts
Howard et al. [41]	USA	Cross-sectional (III)	*n* = 78; M/F = 0/78; mean age = 19.3 ± 3.7 years	Elite athletes (soccer)	Revision = 22; primary = 55	60	BTB autograft, hamstring autograft, quadriceps autograft, and allografts
Koca et al. [42]	Sweden	Cross-sectional (III)	*n* = 156; M/F = 66/90; mean age = 30.1 ± 5.9 years	Recreational athletes (soccer, downhill skiing, handball, basketball, and American Football)	Revision = 78; primary = 78	88	Hamstring and BTB autografts
Lai et al. [43]	Australia	Case series (IV)	*n* = 158; M/F = 158/0; mean age = 23.6 ± 7.4 years	Elite athletes (Australian Rules football)	Revision = 37; primary = 138	NR	Hamstring autograft, BTB autograft, and allografts
Lefevre et al. [44]	France	Cohort (III)	*n* = 552; M/F = 373/179; mean age = 30.2 ± 8.4 years	Elite and recreational athletes (rugby, soccer, basketball, volleyball, skiing, and cycling)	Revision = 55; primary = 497	12	Hamstring and BTB autografts
Lindanger et al. [45]	Norway	Prospective cohort (II)	*n* = 149; M/F = 73/76; mean age = 24.0 ± 15.5 years	Elite and recreational athletes (soccer, handball, and basketball)	Revision = 14; primary = 135	24	BTB autograft
Mai et al. [46]	USA	Cohort (III)	*n* = 344; M/F = 344/0; mean age = 26.9 ± 3.7 years	Elite athletes (American Football, basketball, hockey, and baseball)	Revision = 12; primary = 344	36	NR
Mardani-Kivi et al. [47]	Iran	Cross-sectional (III)	*n* = 124; M/F = 89/35; mean age = 29.4 ± 8.1 years	NR (soccer, basketball, handball, and martial arts)	Revision = 72; primary = 72	72	Hamstring autograft and Achilles tendon allograft
Okoroha et al. [48]	USA	Case-control (III)	*n* = 118; M/F = 118/0; mean age = 25.3 ± 3.2 years	Elite athletes (American Football)	Revision = 23; primary = 95	11	NR
Ortiz et al. [49]	Argentina	Case series (IV)	*n* = 41; M/F = 39/2; mean age = 29.0 ± 8.9 years	Amateur athletes (soccer, running, rugby, tennis, and martial arts)	Revision = 41; primary = 41	69	Hamstring autograft, BTB autograft, and allografts
Pinheiro et al. [50]	United Kingdom	Case-control (III)	*n* = 200; M/F = 200/0; mean age = 24.1 ± 4.2 years	Elite athletes (soccer)	Revision = 18; primary = 200	24	NR

Table 1 (continued)

Study	Country	Study design (level of evidence)	Population characteristics	Level of play (type of sport)	ACL reconstruction type (*n*)	Follow-up period (months)	Graft type
Tomihara et al. [51]	Japan	Case-control (III)	*n* = 66; M/F = 45/21; mean age = 22.8 years ± 4.9 years	National-level and recreational athletes (NR)	Revision = 22; primary = 44	32	BTB and hamstring autografts
Webster et al. [52]	Australia	Case-control (III)	*n* = 128; M/F = 103/25; mean age = 19.8 ± 2.6 years	Youth athletes (Australian Rules football, basketball, soccer, and netball)	Revision = 119; primary = 119	53	Hamstring autograft, BTB autograft, and allografts

ACL, anterior cruciate ligament; BTB, bone-patellar tendon-bone; F, females; M, males; n, number; NR, not reported

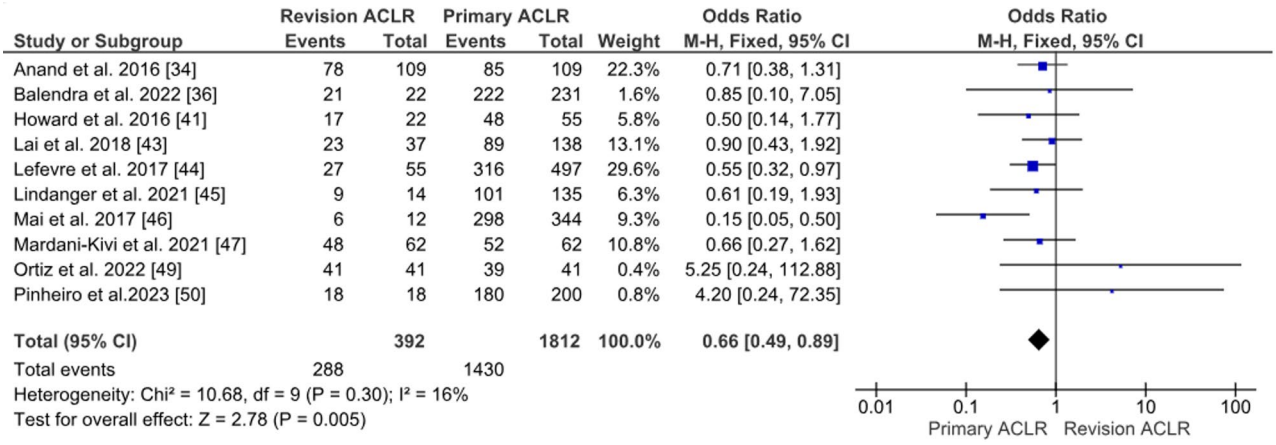


Fig. 2 Overall RTS after revision compared to the primary ACL reconstruction. ACL, anterior cruciate ligament; RTS, return to sport

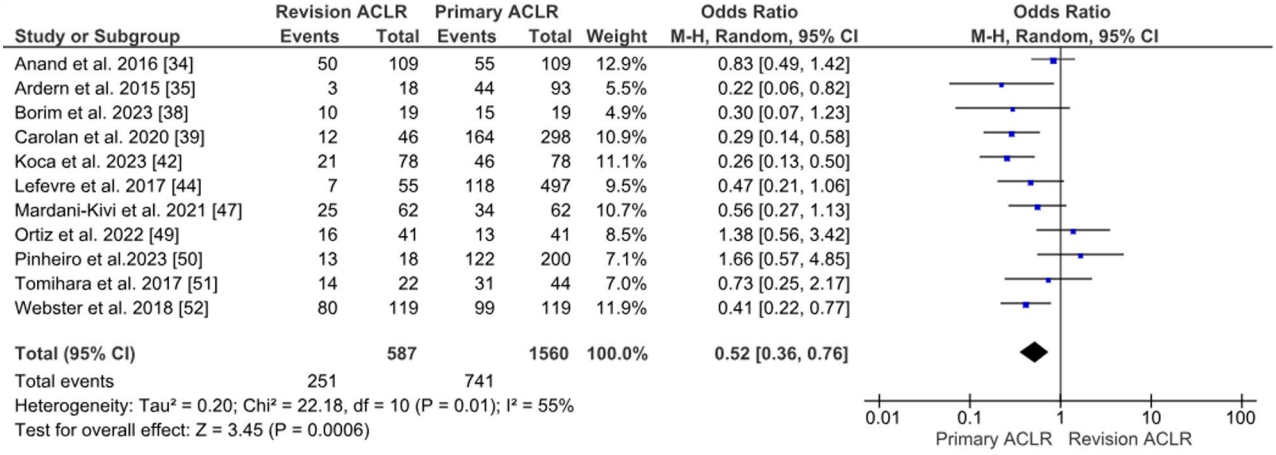


Fig. 3 RTS at the preinjury level of competition after revision compared to the primary ACL reconstruction. ACL, anterior cruciate ligament; RTS, return to sport

Sensitivity analysis induced a negligible increase with respect to the effect size (OR = 0.69 [95% CI 0.50 to 0.95]), while heterogeneity among studies decreased and remained low ($I^2 = 0\%$) (Supplementary file 3).

Eleven investigations addressed RTS at the preinjury level of competition of the examined population (Fig. 3). Athletes who underwent the revision ACL reconstruction had slightly decreased odds of RTS at the preinjury level of competition compared to their counterparts with a history of primary ACL reconstruction (43% vs. 48%, respectively; -5%; OR = 0.52 [95% CI 0.36 to 0.76]; $p = 0.0006$). A high heterogeneity was noted regarding the studies that assessed the mentioned outcome ($I^2 = 55\%$). Sensitivity analysis did not elicit changes with respect to the OR and level of heterogeneity (Supplementary file 3).

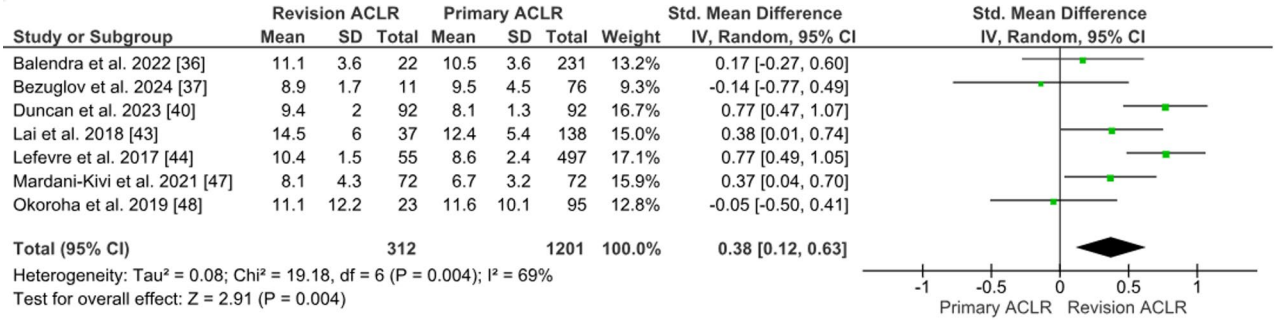


Fig. 4 RTS time after revision compared to the primary ACL reconstruction. ACL, anterior cruciate ligament; RTS, return to sport

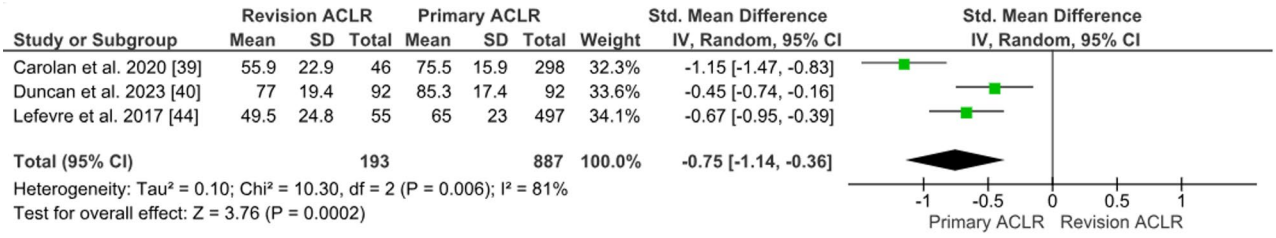


Fig. 5 Psychological readiness to return to sport after revision compared to the primary ACL reconstruction. ACL, anterior cruciate ligament

Seven studies examined RTS time following revision and primary ACL reconstruction (Fig. 4). Athletes who were subjected to revision ACL reconstruction had slightly prolonged RTS time relative to the participants with a history of primary ACL surgery (1.2 months; SMD = 0.38 [95% CI 0.12 to -0.63]; $p = 0.004$). A high level of statistical heterogeneity was recorded concerning the analyzed outcome ($I^2 = 69\%$). Sensitivity analysis increase revealed differences in favor of individuals with a history of ACL revision reconstruction (1.5 months; SMD = 0.43 [95% CI 0.18 to -0.69]), while heterogeneity was only slightly reduced ($I^2 = 69\%$) (Supplementary file 3).

Psychological Readiness to Return to Sport

Three studies explored psychological readiness to return to sport after revision and primary ACL surgery (Fig. 5). Participants with a history of revision ACL reconstruction had moderately lower ACL-RSI scores compared to the athletes who underwent exclusively the primary ACL surgery (-14.2 points; SMD = -0.75 [95% CI -1.14 to -0.36]; $p = 0.0002$). A high heterogeneity was observed among investigations referring to the psychological readiness to return to sport ($I^2 = 81\%$). Sensitivity analysis was not conducted concerning the mentioned outcome.

Quality Assessment

The evaluation of the methodological quality of studies that satisfied all highlighted PECOS requirements is provided in Supplementary file 4. Fifteen studies [34–39, 42, 44, 46–52] had moderate quality with a score between 13 and 18, while 4 investigations [40, 41, 43, 45] had high quality with a score of 19 and 20. The mean quality score for all involved studies was 16.7; hence, the overall methodological quality was moderate. The lowest rated items were "unbiased assessment of the study endpoint," "loss to follow-up less than 5%," and "prospective calculation of the study size." The highest-rated items were "a clearly stated aim," "prospective collection of data," and "contemporary groups."

Certainty of Evidence Assessment

Due to the initial level of evidence in the observational studies, the overall GRADE score for RTS outcomes and psychological readiness to return to sport was very low. Of note, evidence was not downgraded regarding the recorded methodological quality in all relevant parameters. Conversely, evidence was most commonly downgraded due to the high statistical heterogeneity among studies that evaluated primary and secondary outcomes. Publication bias was estimated only for the variables overall RTS and RTS at the preinjury level of competition. No quality of evidence was upgraded with respect to the values of effect size, regardless of whether OR and SMD were taken into account. More details pertaining to the certainty of evidence assessment of each outcome are given in Supplementary file 5.

Discussion

Overall Summary of Evidence

To date, to the best of the authors' knowledge, this is the first investigation that summarized the currently existing body of scientific evidence referring to the comparison between revision and primary ACL reconstruction regarding RTS outcomes and psychological readiness to

return to sport. Namely, consistent with the hypothesis, athletes who were subjected to revision ACL reconstruction were less likely to RTS, had decreased odds of RTS at the preinjury level of competition, had prolonged RTS time, and reduced psychological readiness to return to sport relative to the respondents with a history of primary ACL surgery. More specifically, the obtained results indicated that individuals who underwent revision ACL reconstruction had 6% and 5% reduced rates of RTS and RTS at the preinjury level of participation, respectively, and 1.2 months of extended RTS time compared to their counterparts who experienced only primary ACL reconstruction. Similarly, the present meta-analysis also demonstrated that psychological readiness to return to sport was 14.2 points lower among individuals with a history of revision ACL reconstruction than after primary surgery of the mentioned ligament.

Return to Sport Outcomes

The findings of the present study regarding RTS outcomes are partially consistent with the existing body of knowledge referring to the self-reported outcomes, including subjectively evaluated physical activity, knee function, and quality of life. More precisely, several relevant meta-analyses [53, 54] did not reveal differences between revision and primary ACL reconstruction concerning physical activity levels assessed with the Tegner Activity Scale. It is crucial to highlight that the mentioned scale is quite limited and simple, evaluating only one out of 4 available FITT (frequency, intensity, time, type) components [55]. On the other hand, the RTS outcomes are more complex phenomena that are determined by various biological, psychological, and sociological factors, potentially explaining the emphasized incompatibility. Still, compelling evidence indicated that the values of knee function and quality of life were lower after revision relative to the primary reconstruction of ACL [53, 54]. Most importantly, individuals with a history of revision surgery of the examined ligament had considerably reduced Knee Injury and Osteoarthritis Outcome Score with respect to the sports activities compared to the respondents who were subjected to the primary reconstruction [54]. As previously shown, the current study revealed that revision ACL reconstruction was negatively associated with evaluated RTS outcomes. Still, given the vulnerability of the knees in athletes with multiple ACL tears, the preventative context should also be emphasized for the adequate interpretation and practical implications of the findings. Namely, the rigorous and detailed monitoring of the analyzed population during the overall RTS process is considered crucial. Specifically, although the criteria of the objective tests are fulfilled, including the limb symmetry index of the isokinetic strength of quadriceps and hamstring muscles, and well-known hop tests, the decision-making that pertains to returning to sports activities should be carefully conducted due to the potential risks of subsequent ACL injuries following revision reconstruction. Of note, the available evidence indicates that earlier RTS after ACL reconstruction is associated with further ipsilateral and contralateral ACL injuries, highlighting the role of a 9-month cutoff value [56–58]. Therefore, in addition to the need to increase the overall RTS rate and RTS at the preinjury level of competition among athletes who were subjected to revision ACL reconstruction, the timing of the RTS of 10.8 months could even be prolonged to ensure adequate adaptation of the implemented graft type following surgery [59] and reduce the probabilities of new ACL tears. Overall, exercise specialists and physiotherapists should prioritize an optimal rehabilitation approach to enhance RTS rates and prevent third ACL tears that would negatively impact athletes' longevity and successful participation in high-level sports.

Psychological Readiness to Return to Sport

The current investigation revealed that athletes with a history of revision ACL surgery had 14.2 points lower ACL-RSI scores compared to their peers who were subjected to the primary ACL reconstruction. These differences could be considered clinically relevant, as Webster and Feller previously established thresholds of 13.4 points using the ACL-RSI scale [17]. Moreover, the mean score of psychological readiness to return to sports following revision reconstruction (60.1 points) was significantly below the cutoff values recommended by Webster and Feller necessary for RTS [20]. Hence, the lower psychological readiness to return to sport among respondents with a history of revision ACL reconstruction is reasonable and expected, most likely due to high levels of fear of additional ACL injury, which is a vital component of the mentioned variable in this context. Additionally, the existing literature consistently demonstrated that a greater level of psychological readiness to return to sport was associated with a subsequent ACL injury following a reconstruction procedure [60, 61]. Collectively, only carefully controlled improvements in psychological readiness to return to sport are recommended for athletes who were subjected to revision ACL reconstruction. On the other hand, recent evidence highlighted that low levels of psychological readiness to return to sport were strongly and negatively associated with RTS after revision ACL surgery [62]. Moreover, the effect estimates of the present study highlighted that revision ACL reconstruction was more detrimental to psychological readiness to return to sport than to primary outcomes (moderate vs. trivial effect size, respectively). For all these reasons, the potential inclusion of the sport psychologist in rehabilitation following ACL revision reconstruction should

enable a careful and comprehensive evaluation of the variable of interest. In summary, the administration of physical exercise and psychological interventions could be considered a gold-standard rehabilitation approach for the addressed parameters among athletes with revision ACL reconstruction, accounting for both improvements and prevention. However, further studies should verify the effectiveness of the emphasized practical recommendation for athletes who underwent revision ACL reconstruction.

Strengths and Limitations

The present investigation has numerous strengths that are necessary to emphasize. Namely, it is crucial to highlight that only the findings of the recently available studies were taken into account, providing state-of-the-art evidence with respect to the addressed topic. Moreover, a significant number of studies evaluated RTS rate outcomes; hence, there is clear evidence that ACL revision surgery negatively correlated with the mentioned variables. Additionally, available studies were conducted worldwide, including Australia, Europe, North America, South America, and Asia, potentially improving the generalization of the most relevant findings. Finally, the obtained results have robust practical relevance for professions involved in rehabilitation following revision ACL reconstruction, such as exercise specialists, physiotherapists, and sport psychologists.

Several apparent limitations, particularly concerning methodology, need to be comprehensively analyzed. Most importantly, a substantially lower number of ACL revision reconstructions were observed compared to the primary ACL surgeries. In addition, an inequality referring to sex was also recorded, indicating a paucity of evidence among females with a history of ACL reconstruction. Moreover, the role of the relevant graft types in the context of the revealed differences for all addressed outcomes was not explored. It should also be noted that a moderate methodological quality was observed among the included studies, as well as a very low overall quality of evidence for each analyzed outcome. Further, a high level of heterogeneity was revealed for outcomes such as RTS at the preinjury level of participation, RTS time, and psychological readiness to return to sport. It is necessary to acknowledge that only observational studies were taken into account, most commonly indicating level III and level IV of evidence. Psychological readiness to return to sport was addressed only in three investigations. Therefore, the comparison with respect to the secondary outcome between revision and primary ACL surgery should be interpreted with some caution.

Gaps in the Existing Knowledge

The findings of this systematic review and meta-analysis identified numerous gaps in the existing literature and interesting future directions. Namely, all studies conducted on elite athletes mostly consisted of soccer players. Hence, examining differences in assessed outcomes following revision and primary ACL surgery among high-level basketball, handball, or volleyball players is highly recommended in the future. The current study highlighted that the overall follow-up period was 43 months (3.6 years), mostly including short- and medium-term follow-up assessments. Therefore, more studies exploring primary and secondary parameters long-term following revision and primary ACL reconstruction are needed. Furthermore, in terms of the age of respondents, investigations comparing revision and primary ACL reconstruction regarding each analyzed outcome among athletes under 18 years old are warranted in the future. Most importantly, factors associated with revealed differences are quite unclear. More precisely, the role of variables such as sex, type of sport, level of play, and follow-up period regarding the highlighted disparities in all outcomes needs to be thoroughly examined in the future.

Conclusion

The present study revealed that athletes who were subjected to revision ACL surgery had a 6% lower rate of RTS, a 5% reduced rate of RTS at the preinjury level of competition, and 1.2 months of extended RTS time relative to their peers with a history of primary ACL surgery. In addition, individuals with a history of revision ACL surgery had 14.2 points lower psychological readiness to return to the sport than respondents who underwent the primary reconstruction of the examined ligament. Moreover, the optimal rehabilitation modality, encompassing exercise-based interventions and psychological sessions, is recommended to achieve benefits for the assessed parameters and to prevent subsequent knee injuries. Lastly, more long-term follow-up studies with respect to all evaluated parameters are highly warranted in the future.

Abbreviations

ACL	Anterior cruciate ligament
ACL-RSI	Anterior cruciate ligament return to sport after injury
CI	Confidence interval
GRADE	Grading of recommendations assessment, development, and evaluation
MINORS	Methodological index for non-randomized studies
OR	Odds ratio
PRISMA	Preferred reporting items for systematic reviews and meta-analyses
RTS	Return to sport
SMD	Standardized mean differences

Supplementary Information

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Supplementary Material 1.

Supplementary Material 2.

Supplementary Material 3.

Supplementary Material 4.

Supplementary Material 5.

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Data Availability
Data are available upon reasonable request to the corresponding author.

Declarations

Ethics Approval and Consent to Participate
Not applicable.

Consent for Publication
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Competing Interests
The authors declare the absence of competing financial interests or any personal relationship that could influence the content available in the manuscript.

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